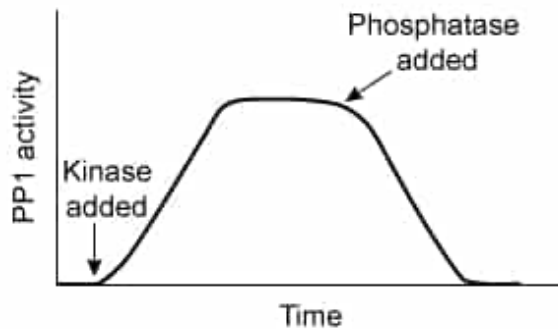


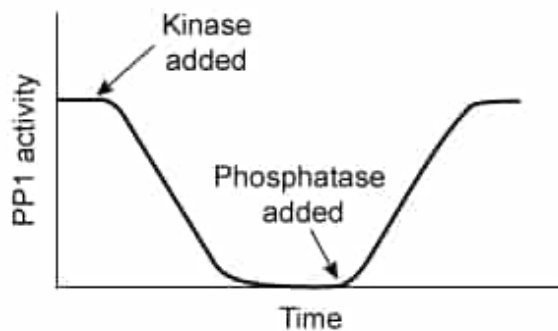
Protein phosphatase 1 (PP1) removes phosphate groups from target proteins. A researcher hypothesizes that PP1 can be inactivated by phosphorylation. Which graph depicts experimental results that would best support this hypothesis? (Assume that PP1 initially has no phosphate groups and that any enzymes added act on PP1 to the greatest extent possible.)

☐ A.



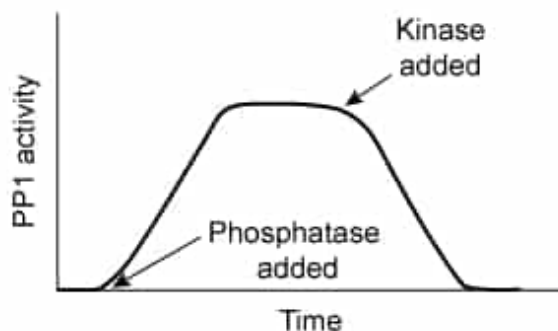
[11%]

☒ B.



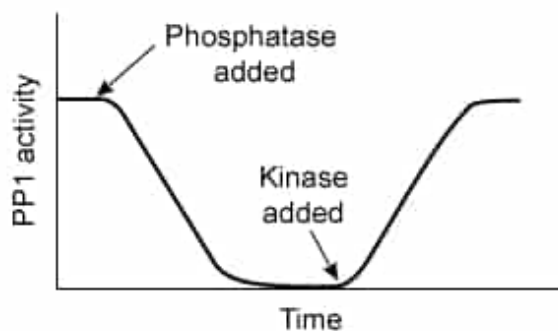
[59%]

☐ C.



[21%]

☐ D.



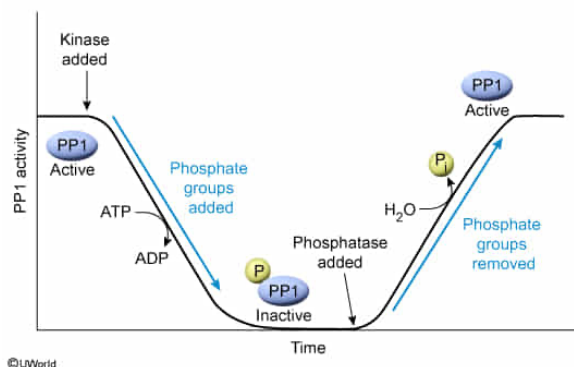
[7%]

Correct

59% answered correctly

Explanation:

Phosphorylation and enzyme activity



Kinases are enzymes that transfer a phosphate group from a **nucleoside triphosphate** (usually ATP) to a target molecule. Although some kinases (eg, **pyruvate kinase**) facilitate the reverse reaction, often the reverse reaction is too **energetically unfavorable** to occur. Consequently, dephosphorylation of a target molecule is typically catalyzed by a **phosphatase**, which **hydrolyzes** the bond between the phosphate group and the molecule, producing the dephosphorylated substrate and inorganic phosphate (P_i).

Many enzymes are **regulated by phosphorylation and dephosphorylation**. **Some enzymes are activated** by phosphorylation while **others are inactivated**. In the question, the researcher hypothesized that PP1 is *inactivated* by phosphorylation. If this is correct, then the addition of a **kinase will phosphorylate PP1** and cause a **reduction in PP1 activity**. Later addition of a **phosphatase will remove the added phosphate** and thereby **reactivate PP1**, causing an **increase in its observed activity**. Of the given choices, only the graph in Choice B correctly depicts the expected effects of a kinase and a phosphatase on PP1.

(Choice A) This graph shows an *increase* in activity when a kinase is added and a decrease in activity when a phosphatase is added. This result would indicate that PP1 is *activated* by phosphorylation, not inactivated.

(Choice C) This graph would be consistent with the given hypothesis if PP1 were initially in the phosphorylated state, but the question specifies that PP1 *initially has no phosphate groups*. Therefore, the addition of a phosphatase at the beginning of the assay should have no effect because there are no phosphate groups to remove.

(Choice D) Adding a phosphatase at the beginning of the assay should have no effect because PP1 initially has no phosphate groups to remove. In addition, if phosphorylation inactivates PP1, the later addition of a kinase should result in *decreased* activity, not *increased* activity.

Educational objective:

Kinases transfer phosphate groups from nucleoside triphosphates (usually ATP) to target molecules. For most kinases, the reverse reaction is unfavorable, so phosphate groups are instead removed through hydrolysis by phosphatases. Phosphorylation may either activate or inactivate an enzyme.

Time spent:0 sec

QID:404168

Subject:
Biochemistry

System:
Enzymes